Real-World Problems and Significant Digits

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 4 — Real-World Problems and Significant Digits	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.6

2. Standards

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must. Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not. The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 2, 6 and 7 — the last move on every board is asking how much of the answer the measurements actually support, before the board goes up.
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Every problem from Notes II on — each number is a measured quantity with a unit, because the rule has no meaning without one.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 2.6×10^5 and 2.61×10^5, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Essential question. How much of an answer is worth writing down, when the numbers that produced it were measured?

Building on. 4.01–4.03 — all four operations in scientific notation; Grade 5 and 6 — place value and rounding to a named place.

Working toward. 4.05–4.08 — the order of operations with exponents and radicals; MA.912.GR.4 — measurement and precision in geometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why does the last digit of a measurement carry less information than the digits before it?	DOK 2
A sum of two two-digit measurements can have three significant digits. How is that possible, when a product of the same two could only have two?	DOK 3
Ellis uses the product rule on a sum and throws away two digits that were measured. What is each of the two rules actually protecting?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	How many significant digits are in 0.00470 ?	***3***	2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]
2	A pond's surface measures 1.4×10^2 m by 8.5×10^1 m. The pond is 3 meters deep. Find the surface area. Round to the correct number of significant digits.	***1.2×10^4 square meters***	anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^4 — the arithmetic without the rounding [8.NSO.1.6 clarification]
3	Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5. How should the answer be written?	***7.41×10^5***	7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^5 — no rounding at all
4	$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured. Write the sum with the correct precision.	***3.0×10^3 m***	2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9×10^3 — 2,971 rounded down instead of up [rounding to the hundreds place]
5	How many significant digits are in 0.03080?	***A — 4***	(B) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]
6	A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits?	***B — 2.6×10^5 cm^2***	(A) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one]; (D) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification]
7	A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits?	***1.4×10^2 km/h***	140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low
8	A measured length times a measured width gives an area of 3.6×10^3 m^2. The length measured 9.0×10^1 m. What did the width measure, to two significant digits?	***4.0×10^1 m***	4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high
9	Ellis says that because 9,145 has 4 significant digits and 4.8 has 2, the sum 9,145 + 4.8 must be written with 2 significant digits, as 9.1×10^3. Explain why he is wrong, and give the correct sum.	***9,150 = 9.150×10^3***	'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 9,149.8 — no rounding at all, which claims a tenth that 9,145 never measured

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the four counting rules as a card with one worked number each, in the order of Notes I, and let it stay on the desk through the round. On every problem the student marks the significant digits on the two measurements before doing any arithmetic, so the count is decided before it is needed.

ELL. Significant, precise and measurement are three words doing specific work today. Put MEASUREMENT = a number someone measured beside COUNT = a number someone counted, with 4.20 meters under one and 24 students under the other, and ask which kind each number is before every problem.

Enrichment. Ask what 1,000 means when it is written with no decimal point — one significant digit or four — and how a scientist would write each of the two. Then ask why the state guide's own Puerto Rico item is genuinely ambiguous.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say what the rule for a sum actually is, not only that Ellis used the wrong one, and give 9.150×10^{3}. Read the boards, not the papers.

Vocabulary. **significant digits**; **leading zero**; **measurement**

Materials. Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 4 test: Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S.